
Research Copilot

Software Requirements Specification

Submitted By

Duaa Siraj (24K-0524)

Ayra Omar (24L-0628)

Asra Zehra (24K-0667)

Ruhab Ayesah Farooq (24K-0537)

Submitted To

Miss Maqsood Khatoon

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1. Introduction

1.1 Problem Description

Problem Statement:

There has been an exponential increase in volume of academic publishing over the last few years, with millions being published annually across thousands of journals. The primary medium for consuming this information remains the static PDF.

The current manual researching system makes it time-prohibitive to stay organized and on-track. Needing multiple tabs and having to decipher the meaning of every second word makes it difficult to actually get work done.

Key Deficiencies:

- **Manual Latency:** Readers must manually cross-reference data and citations in an attempt to ensure relevance and credibility. This takes a copious amount of time that could be used more productively.
- **Lack of Accessibility:** Someone who isn't as well-versed in the technicalities of the study finds it tedious to comprehend what they are reading because of the jargon and structural density.

These setbacks only act as demotivators for those looking to get started with research. In the current educational landscape where the literacy crisis is at an all time high, it is essential to make academic material accessible to everyone and encourage the democratization of knowledge.

1.2 Proposed Solution

Research Copilot is a digital assistant designed to make studying academic papers faster and much easier for everyone. Instead of just reading a standard, static PDF, users can work in a dynamic space that automatically pulls out the most important information. The software works by taking an uploaded paper and breaking it down using AI to find things like the main findings, the methods used, and any potential flaws or concerns.

At the same time, the system searches online databases to find other studies that either agree with, disagree with, or are simply related to the paper being read. These other studies are organized into clear categories so the user can quickly check if the paper's claims are backed up by other researchers. The system also creates a visual map of citations, making it easy to see how different research papers are connected to one another.

To help with difficult or confusing language, there is a built-in chatbot that remembers the paper's details and can answer specific questions or explain hard-to-understand terms in simple English. By handling the tedious parts of searching and organizing, Research Copilot lets the user focus on actually understanding the ideas and thinking critically about the research.

1.3 Scope

Similar Pre-existing Systems

SciSpace: SciSpace is one of the most comprehensive AI powered academic platforms available. It provides a conversational Copilot feature, PDF uploads, literature review, and access to millions of research papers. It differs from Research Copilot in how it structures and presents data; where research copilot provides a tri-categorised breakdown for the uploaded research paper, SciSpace provides the same information in one single interface.

Consensus: Consensus is a system that takes in a reserch question, and returns a summary of how published research papers answer that question. It differs from Research Copilot by focusing on the question, where Research Copilot focuses on the paper itself.

Connected Papers: Connected Papers is a tool used to find similar papers to a source paper. Connected Papers only provides a visualization tool, not an analysis of the paper the way Research Copilot does.

NotebookLM by Google: NotebookLM allows users to upload sources (books, research papers, articles, etc.) and interact with them conversationally or generate supporting material such as slides, flashcards, graphs etc. It differs from Research Copilot by being a more generic academic tool, rather than being aimed specifically towards research papers the way Research Copilot is. It also does not provide any statistics or supporting/conflicting information regarding uploaded sources.

Inclusions

- Accept a user-uploaded research paper
- Generate a comprehensive analysis of the paper which includes key findings, concerns, limitations, statistical models, methodologies, etc.
- Collect Related Papers and categorize them into Supporting studies, Conflicting studies, and Related studies
- Provide a conversational Chatbot to answer questions using the research papers as a source

- Provide a citation graph that illustrates the references cited within the uploaded paper
- Provide an accessible UI with dark mode and a multi-panelled view

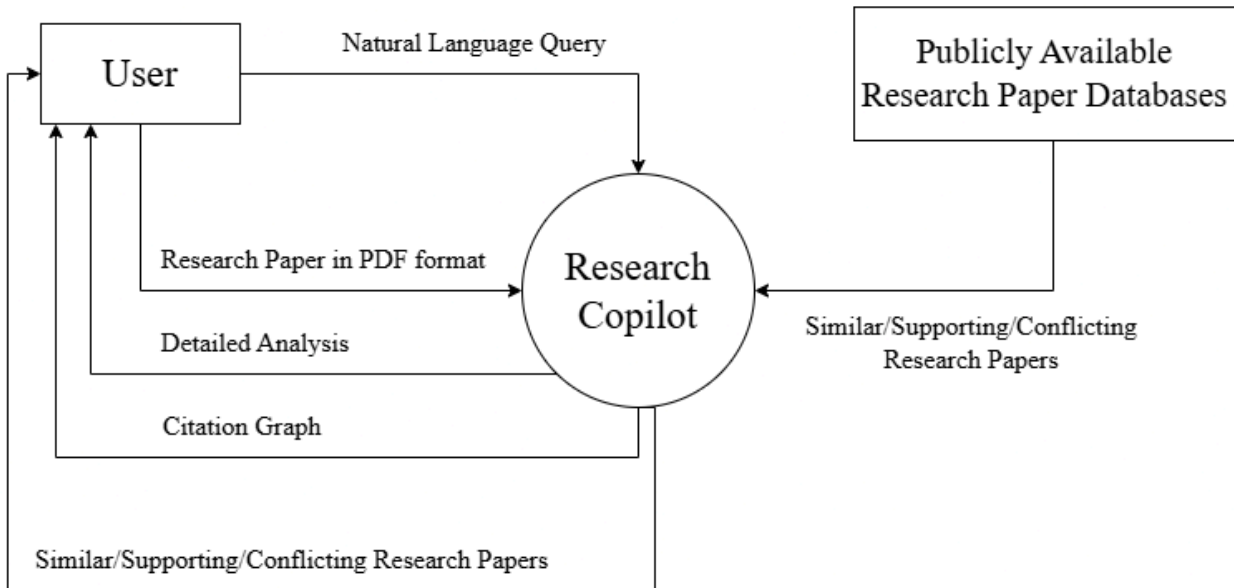
Exclusions

- A database to store citations
- Third party integrations
- Academic writing assistance
- Analysis of multiple papers simultaneously
- Offline usage

1.4 Objectives

- **Automated Paper Analysis:** The system will accept a research paper in PDF format and automatically extract and present structured metadata, including the paper's title, concise summary, key findings, identified limitations and concerns, statistical models employed, and population or sample characteristics.
- **Literature Context Discovery:** The system will locate and organize related academic literature by categorizing retrieved studies as supporting, conflicting, or related to the uploaded paper, enabling users to validate claims without manual cross-referencing.
- **Contextual chatbot:** The system will provide a conversational interface to ask queries relating to the uploaded paper
- **Citation Graph visualization:** The system will render an interactive citation graph with clickable links(if available).
- **User Profile and Paper History:** The system will support authenticated user accounts with the ability to save uploaded papers, analyses, and summaries for future retrieval.

1.5 Context Diagram



1.6 Use Case Diagram

